TEST CODE **01212032**

# FORM TP 2013006

JANUARY 2013

## CARIBBEAN EXAMINATIONS COUNCIL

## CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
## EXAMINATION

## CHEMISTRY

### Paper 032 – General Proficiency

### Alternative to SBA

***2 hours and 10 minutes***

**READ THE FOLLOWING INSTRUCTIONS CAREFULLY.**

1. Answer ALL questions in this booklet.
2. Use this booklet when responding to the questions. For EACH question, write your answer in the space indicated and return the booklet at the end of the examination.
3. You may use a silent, non-programmable calculator to answer items.
4. You are advised to take some time to read through the paper and plan your answers.



**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.**

**Answer ALL questions.**

**1.** The students in a class conducted an experiment to determine the heat of neutralization for the reaction of hydrochloric acid with sodium hydroxide.

The materials, reagents and the procedure of the experiment are given below.

MATERIALS: Styrofoam cup with cover; thermometer; balance; 2 beakers; 2 measuring cylinders; stirrer.

REAGENTS: 2 mol $dm^{-3}$ $HCl$; 2 mol $dm^{-3}$ $NaOH$.

PROCEDURE:

1. Place 150 ml each of the HCl and NaOH into separate labelled beakers and let stand for approximately 5 minutes.
2. Weigh, together, the Styrofoam cup with cover, thermometer and stirrer, and record the mass.
3. Measure out **exactly** 50 ml of 2 mol $dm^{-3}$ $HCl$ and transfer it as completely as possible to the Styrofoam cup.
4. Cover the cup and record the temperature.
5. Measure out exactly 50 ml of 2 mol $dm^{-3}$ $NaOH$ and transfer it as completely as possible to the Styrofoam cup.
6. Replace the lid and stir the solution gently. Record the **highest** temperature reached.
7. Weigh the Styrofoam cup with cover, thermometer, stirrer and solution; record the mass.

RESULTS:

Figure 1 shows the results of the experiment.

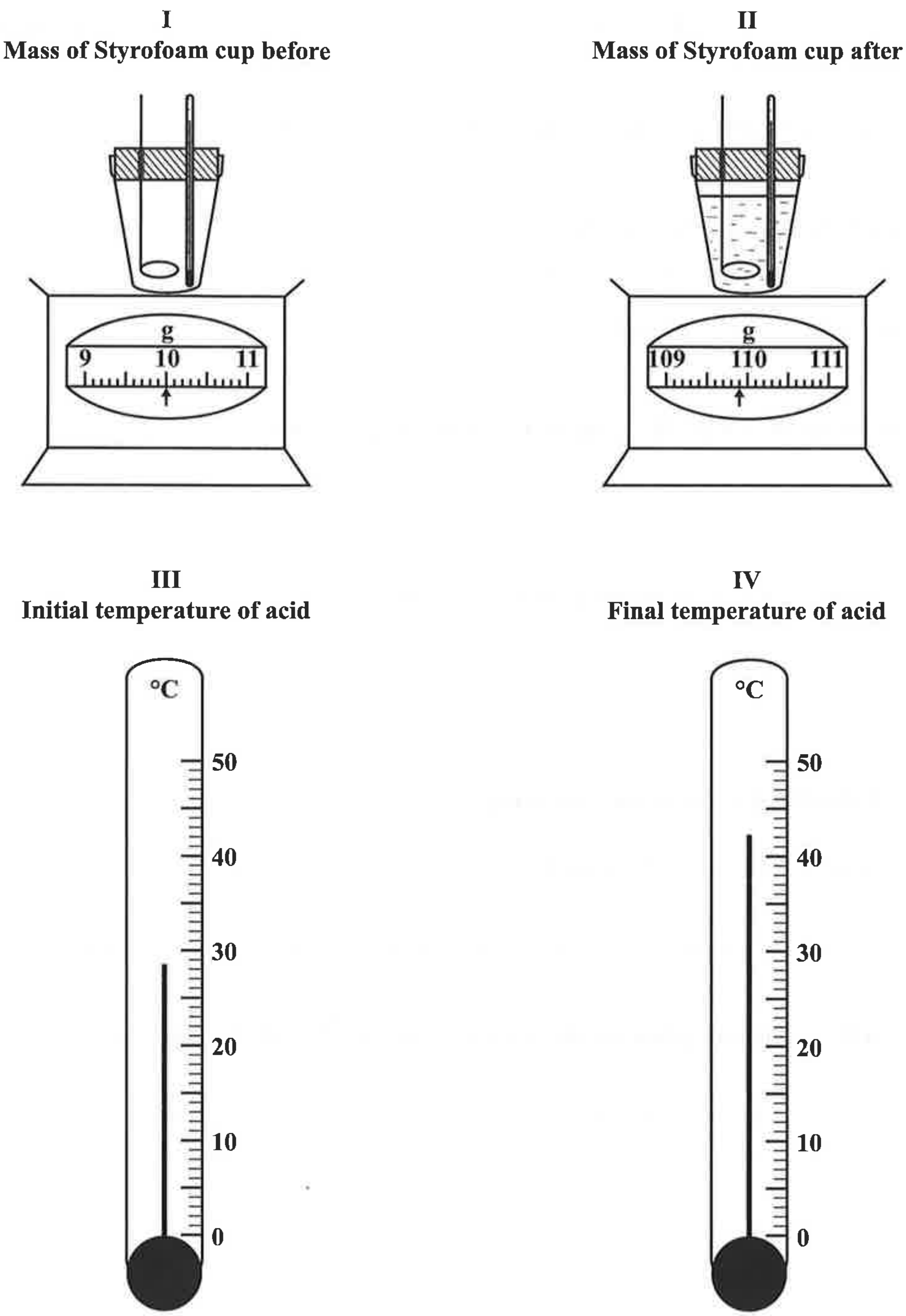


**Figure 1. Results of experiment**

(a) Use the information given in Figure 1 to complete Table 1.

**TABLE 1: RESULTS FROM EXPERIMENT**

| Measure | Reading |
|---|---|
| (i) Mass of empty Styrofoam cup, lid, thermometer and stirrer | |
| (ii) Temperature of acid before reaction | |
| (iii) Highest temperature reached | |
| (iv) Mass of Styrofoam cup, lid, thermometer, stirrer and solution | |

**(6 marks)**

(b) Write a chemical equation for the reaction.

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**(1 mark)**

(c) Calculate EACH of the following:

(i) The mass of solution

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**(1 mark)**

(ii) The temperature change that occurred during the reaction

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**(1 mark)**

(iii) The heat change for the reaction.

[Heat change = mass of solution × 4.2 J $g^{-1}$ $°C^{-1}$ × temperature change]

**(1 mark)**

(iv) Moles of water produced

**(2 marks)**

(v) Heat of neutralization

**(1 mark)**

(d) State whether neutralization reactions are endothermic or exothermic.

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**(1 mark)**

(e) State THREE precautions that should be taken during the experiment.

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**(3 marks)**

(f) State THREE possible sources of error.

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**(3 marks)**

(g) If the students had used a beaker instead of a Styrofoam container, would you expect the experimental heat of neutralization to increase, decrease, or remain the same? Briefly explain.

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**(2 marks)**

(h) Calculate the mass of NaOH pellets required to make 100 ml of a 2 mol $dm^{-3}$ solution of NaOH.

**(2 marks)**

(i) A stirrer made from metal ONLY should not be used in this experiment. Suggest how such a stirrer could be adapted for use in the experiment.

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**(2 marks)**

**Total 26 marks**

**2.** A student conducted a number of tests on a solid mixture, X. The inferences made are recorded in Table 2. Complete Table 2 by filling in the observations based on the inferences made.

**TABLE 2: TESTS ON MIXTURE X**

| Test | Observation | Inferences |
|---|---|---|
| (a) Distilled water was added to a portion of X and the resulting mixture stirred and filtered. (The residue was set aside for use later.) The filtrate was divided into 3 equal portions and tests (b) to (d) **done on separate portions.** | | |
| (b) Dilute nitric acid followed by a few drops of silver nitrate solution was added. Ammonium hydroxide solution was added to the resulting mixture. | •<br>•<br>•<br>**(3 marks)** | $Cl^-$ ions are present. |
| (c) A few drops of potassium iodide solution were added. | •<br>**(2 marks)** | An oxidizing agent is present. |
| (d) Aqueous sodium hydroxide was added until in excess. | •<br>•<br>**(2 marks)** | Iron(III) ions are present. |
| (e) Dilute nitric acid was added to the residue from test (a) and the gas produced passed through lime water. | •<br>•<br>•<br>**(3 marks)** | Carbonate ions are present. |

**Total 10 marks**

GO ON TO THE NEXT PAGE

**3.** You are provided with three colourless compounds – *anhydrous calcium nitrate, anhydrous calcium carbonate* and *calcium hydroxide* – which are placed in unlabelled containers. You are required to plan and design an experiment using the 'action of heat' to distinguish these three compounds.

Hypothesis: The compounds, *anhydrous calcium nitrate, anhydrous calcium carbonate* and *calcium hydroxide*, can be distinguished by the application of heat.

Your answer should include the following:

(a) Apparatus and materials

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**(2 marks)**

(b) Procedure

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**(3 marks)**

(c) State ONE precaution that should be taken in conducting the experiment.

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**(1 mark)**

(d) State the expected observations for this experiment in a fully labelled table.

**(3 marks)**

(e) Discuss the observations in the table, stating clearly how they can be used to distinguish the three compounds.

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**(3 marks)**

**Total 12 marks**

**END OF TEST**

**IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.**